(4) Quotients of polynomials namely rational functions of the form

$R(x) = \dfrac{p(x)}{q(x)}$ , are continuous at every point where $q(x) \neq 0$, and

$$\lim_{x \to x_0} R(x) = \lim_{x \to x_0} \frac{p(x)}{q(x)}$$

$$= \frac{\lim\limits_{x \to x_0} p(x)}{\lim\limits_{x \to x_0} q(x)} = \frac{p(x_0)}{q(x_0)} = R(x_0) .$$

(5) The circular functions sin $x$ and cos $x$ are continuous at every point of their domain $\mathbb{R} = (-\infty, \infty)$ since $\lim\limits_{x \to x_0} \sin x = \sin x_0$, $\lim\limits_{x \to x_0} \cos x = \cos x_0$ .

As a consequence, tan $x$, cot $x$, cosec $x$, sec $x$ are continuous on their proper domains in view of the reciprocal and quotient rules in the algebra of limits.

(6) The $n$th root functions, $f(x) = x^{\frac{1}{n}}$ are continuous in their proper domain since $\lim\limits_{x \to x_0} \left( x^{\frac{1}{n}} \right) = x_0^{\frac{1}{n}}$.

(7) The reciprocal function $f(x) = \dfrac{1}{x}$ is not defined at 0 and hence it is not continuous at 0. It is continuous at each point of $\mathbb{R} - \{0\}$ .

(8) $h(x) = \begin{cases} x+1, & x \leq 0 \\ x^2+1, & x > 0 \end{cases}$

The domain of $h$ is all of real numbers and

$$\lim_{x \to 0^-} h(x) = \lim_{x \to 0^-} (x+1) = 1 = h(0)$$

$$\lim_{x \to 0^+} h(x) = \lim_{x \to 0^+} (x^2+1) = 1 = h(0) .$$

Thus $h(x)$ is continuous at $x = 0$.

Indeed, $h(x)$ is continuous at each point of $(-\infty, 0)$ and each point of $(0, \infty)$ and hence $h$ is continuous in the whole of $(-\infty, \infty)$.

(9) The greatest integer function $f(x) = \lfloor x \rfloor$ is not continuous at $x = 0$.

For, $\lim\limits_{x \to 0^-} \lfloor x \rfloor = -1$ and

$$\lim_{x \to 0^+} \lfloor x \rfloor = 0$$

It is discontinuous at each integer point. In fact,

$$\lim_{x \to n^-} \lfloor x \rfloor = n - 1 \text{ and}$$

$$\lim_{x \to n^+} \lfloor x \rfloor = n.$$



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